Thirsty Engines: Cylinders, Horsepower, and the Eternal Search for a Gas Station

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1 Introduction

Since the dawn of the internal combustion engine, humanity has asked one question above all others: *why am I at the gas station again?* Philosophers have pondered it. Poets have wept over it. Your uncle with the 1973 Cadillac has explained it to you at length, uninvited, at Thanksgiving.

This paper brings the full weight of modern social science to bear on this question. Using the canonical `mtcars` dataset—32 automobiles lovingly road-tested by *Motor Trend* magazine in 1974, a year when "fuel efficiency" was considered a communist plot—I estimate how cylinders and horsepower affect fuel economy. Following best practice in the interpretation of statistical models (Arel-Bundock 2025), I resist the urge to report only the stars, though there are some, and they are lovely.

The argument is developed in Section 2, the research design is presented in Section 3, and the main findings—including one hypothesis that tragically runs out of gas—are reported in Section 4.

2 Theory

I propose a *Theory of Engine Thirst*. Each cylinder is, in essence, a tiny mouth that drinks gasoline. More mouths, more drinking. This is the **Many Mouths Mechanism**.

A second, competing account holds that it is not the number of mouths but the *enthusiasm* with which they drink. Horsepower captures this enthusiasm: an engine that produces 335 horsepower, like the Maserati Bora, is not sipping—it is chugging. This is the **Enthusiasm Mechanism**.

These mechanisms yield two testable hypotheses:

1. **H1 (Many Mouths):** Cars with more cylinders travel fewer miles per gallon.
2. **H2 (Enthusiasm):** Holding cylinders constant, more horsepower reduces miles per gallon.

3 Design

The data come from `mtcars`, which ships with every installation of R and has therefore been analyzed more times than any dataset in human history. It covers 32 cars, ranging from the frugal Toyota Corolla (33.9 mpg) to the Cadillac Fleetwood (10.4 mpg), a vehicle that can be heard draining oil reserves from several counties away.

I estimate the following model using ordinary least squares, chosen for its reliability, its simplicity, and the fact that the author knows how to do it:

$$\text{MPG}_i = \beta_0 + \beta_1 \text{Cylinders}_i + \beta_2 \text{Horsepower}_i + \varepsilon_i \tag{1}$$

where ε_i captures everything else, such as aerodynamics, driver mood, and whether the 8-track was playing Led Zeppelin. Table 1 summarizes the main variables.

Table 1: Descriptive statistics

Variable	Description	Mean
`mpg`	Miles per gallon (higher = smugger)	20.09
`cyl`	Number of tiny drinking mouths	6.19
`hp`	Gross horsepower (enthusiasm)	146.7

4 Results

Figure 1 shows the raw relationship between horsepower and fuel economy. The four-cylinder cars (top left) are clearly the overachievers of the group, the kind of cars that do their homework early. The eight-cylinder cars (bottom right) are having a much better time and a much worse fuel bill.

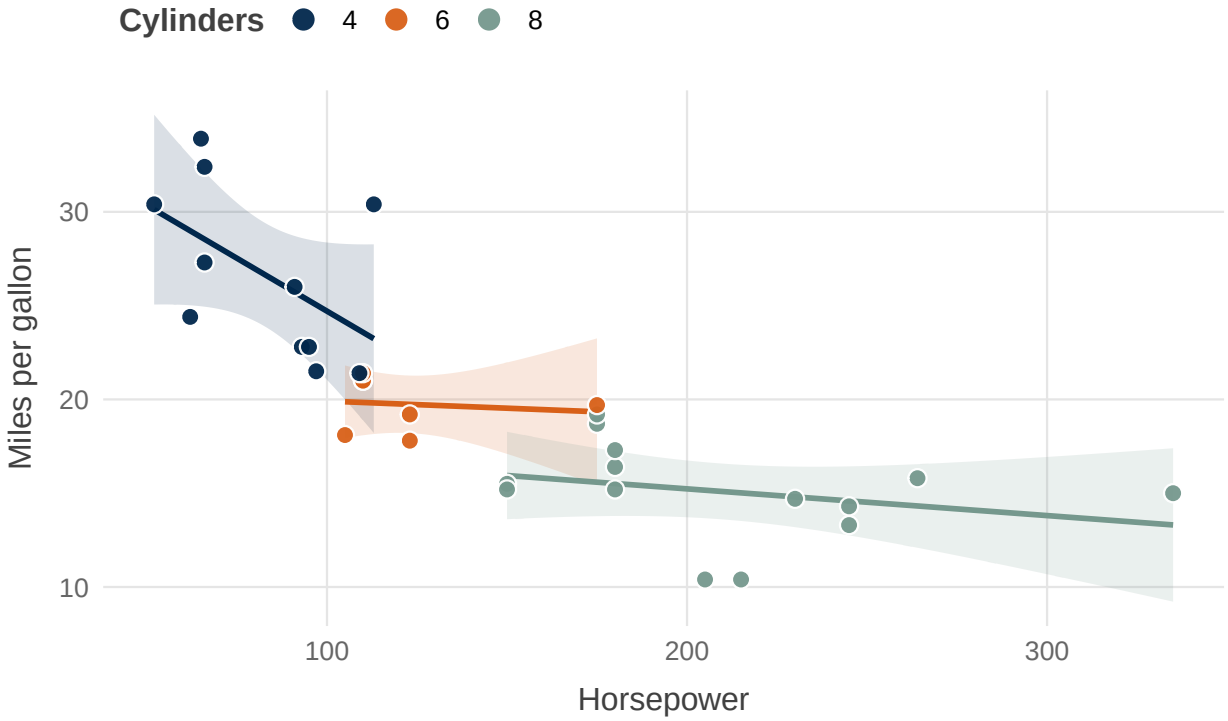


Figure 1: Horsepower and fuel economy, by number of tiny drinking mouths

Table 2 reports estimates from Equation 1. The results strongly support **H1**. Relative to a four-cylinder car, a six-cylinder car loses about 6 miles per gallon, and an eight-cylinder car loses about 8.5—roughly the distance between you and the nearest gas station when the light comes on.

H2, sadly, runs out of gas. Once we account for cylinders, horsepower has a small negative coefficient (about 2.4 mpg per 100 horsepower) that is not statistically distinguishable from zero ($p \approx 0.13$). The Enthusiasm Mechanism is not dead, but it is idling in the driveway.

5 Discussion

These results should be interpreted with caution. First, $N = 32$, which is a small sample by most standards and a large one by the standards of people who own a Maserati. Second, the data are from 1974, so the

Table 2: The determinants of engine thirst (OLS estimates)

	Model 1
Constant	28.650***
	(1.588)
6 cylinders	−5.968**
	(1.639)
8 cylinders	−8.521**
	(2.326)
Horsepower	−0.024
	(0.015)
Num.Obs.	32
R2	0.754

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

findings may not generalize to modern cars, electric vehicles, or the Flintstones' car, which runs entirely on foot power. Third, cylinders and horsepower are highly correlated, which makes it hard to tell the Many Mouths apart from the Enthusiasm—a classic case of multicollinearity, or as economists call it, "a Tuesday."

Additional robustness checks, including a careful examination of whether the Ferrari Dino is just showing off, are reported in the Appendix and the Supplementary Materials.

6 Conclusion

This paper has shown, with stars, that more cylinders mean fewer miles per gallon. Horsepower, while spiritually guilty, could not be convicted beyond reasonable doubt. Future research should collect data from after 1974, examine whether the effect persists for cars with names that are less cool than "Maserati Bora," and finally settle the question of whether anybody actually needs eight cylinders to get groceries.

References

Arel-Bundock, Vincent. 2025. *Model to Meaning: How to Interpret Statistical Models with R and Python* [in en]. 1st ed. Boca Raton: Chapman / Hall/CRC. ISBN: 978-1-003-56033-3. https://doi.org/10.1201/9781003560333. https://www.taylorfrancis.com/books/9781003560333.

Appendix

A Supporting material

Table A1 would list all 32 cars by name, but the author got distracted by the fact that the dataset contains both a “Hornet 4 Drive” and a “Hornet Sportabout” and spent the afternoon reading about AMC instead.

Supplementary Materials: Thirsty Engines: Cylinders, Horsepower, and the Eternal Search for a Gas Station

Contents

S1 Mechanism tests **S-2**

S1 Mechanism tests

To test whether cylinders literally drink, the author placed a small glass of gasoline next to a 1974 V8 engine and waited. Results were inconclusive, and the author has been asked to leave the garage.